High internal accuracy, chance-level external performance

A preregistered external validation of NSGA-II electrode selection for EEG-based detection of mild cognitive impairment (MCI)

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Preregistered on OSF
osf.io/4enh8

75.1% → 45.8%
internal accuracy of the replication → balanced accuracy in 34 new people

0 of 5
methods beat chance in the preregistered external test

94.4%
accuracy in telling which dataset a person comes from, from EEG alone

1 · Background

- **MCI** often precedes dementia. Resting EEG is cheap and portable; a device with **few electrodes** would ease screening.
- **Aljalal et al. (2024, Sci Rep)** reported up to **95%** accuracy with 5–7 electrodes chosen by **NSGA-II**, in 24 people, with LOSO validation.
- But electrodes were **chosen and evaluated on the same people**, no code was released and there was no external test.

2 · Question

> Does that 95% describe how the method would work on **new people**, or how well it fitted the 24 it already had?

Data (public, Isfahan)

	EEG_1	EEG_2
Role	development	single test
People	24 (11 MCI)	34 (18 MCI)
Equipment	19 channels, 256 Hz	same
Analysed	first 10 min	first 10 min

3 · Design

EEG_1 · 24 people
independent Python replication + diagnostic controls

↓

Freeze 5 methods
R (replication) · M0 · M4 · M1 · M6 + physical montage

↓

OSF preregistration
hypotheses, decision rules, SHA-256 of code and data

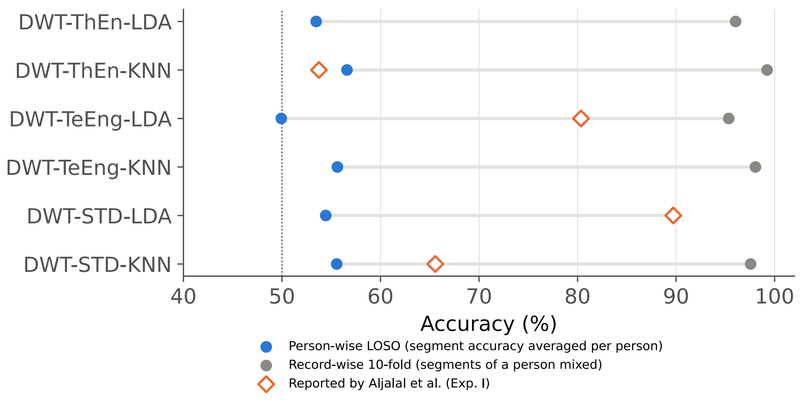
↓

EEG_2 · 34 people
same centre and equipment; single test, no tuning

4 · Methods compared

R	Paper replication: DWT + entropy, 19 channels, NSGA-II, highest-accuracy rule
M0	ICA-cleaned EEG, no Fp1/Fp2/F7/F8, relative band power, regularised LDA
M4	As M0, one electrode per zone (exhaustive search, 180 combinations)
M1	As M0, NSGA-II with the paper's rule (5 seeds)
M6	As M0, binary PSO penalising each electrode (5 seeds)
Physical	M0 and M4 without average reference or ICA: real electrodes

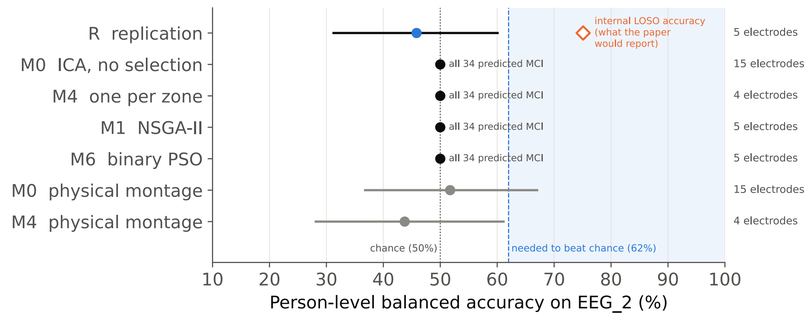
5 · Development: where high figures come from



Same signal and classifiers: person-wise (blue) vs mixed segments (grey); diamonds: values reported by the paper.

- **Segment leakage:** person-wise, 50–57%; mixing segments of the same person, 90–99%. The paper's 89.7% matches our mixed 89.8%.
- **Selection bias:** NSGA-II reaches **72–79%** with random labels.
- **Eyes:** Fp1 and F8, among the paper's most selected, carry 68% and 39% ocular signal.

6 · Preregistered external test



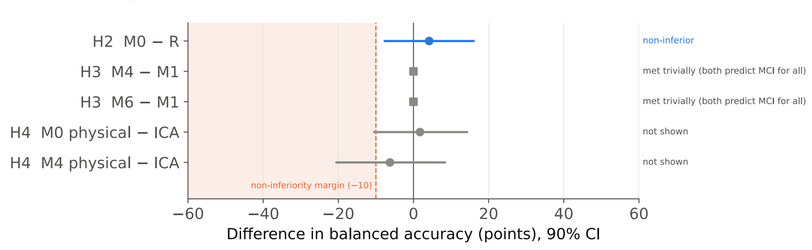
Balanced accuracy on EEG_2 with 95% CI. Blue area: what was needed to beat chance.

The replication had 75.1% internally and scored 45.8% on new people. The other four methods classified **all 34 people as MCI**.

7 · Hypotheses

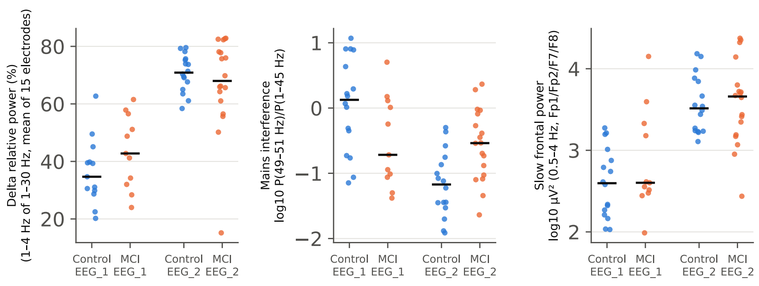
H1	Replication falls below its internal accuracy	Supported
H2	M0 not inferior to the replication	Trivial
H3	M4/M6 not inferior to M1 with fewer electrodes	Trivial / no
H4	Physical montage not inferior to ICA	Inconclusive

"Trivial": met by the rule, but both methods predict MCI for everyone, so it does not show that they work.



Paired differences with 90% CI. Non-inferiority margin: −10 points (orange line).

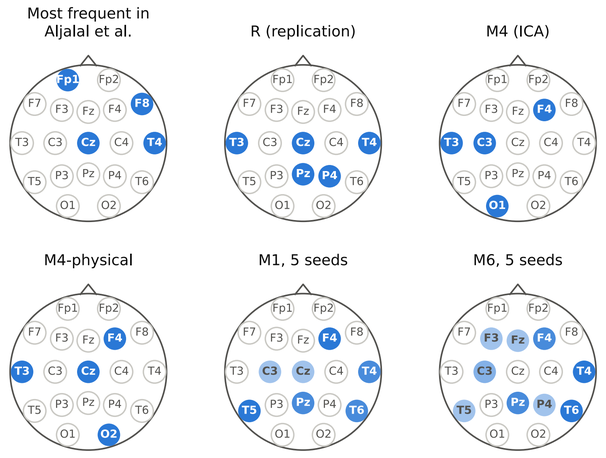
8 · Why did they fail?



Each dot is a person. Differences between datasets exceed differences between groups.

- Both datasets come from the same centre, but **EEG_2 is delta-dominated** (69% vs 40%). Since MCI had slightly more delta in EEG_1, the models send everyone to MCI.
- **Mains interference (50 Hz)** is associated with the diagnosis **in opposite directions** in each dataset: an acquisition confounder.
- Neither the AUC (0.40–0.52) nor standardising each dataset separately revealed a signal.

9 · Selected electrodes



Stable across seeds, but useless on new people: stability is not validity.

10 · Conclusions

- The reported accuracies **do not transfer** to new people.
- **Dataset shift** and **acquisition confounders** outweigh the MCI signal.
- Minimising electrodes only makes sense **after** showing that the signal generalises.

11 · Minimum protocol we propose

- ✓ Evaluate by person, never by segment
- ✓ Compare against shuffled labels
- ✓ Check eyes and mains noise by group
- ✓ Test whether datasets can be told apart
- ✓ Nested selection inside validation
- ✓ Report AUC as well as accuracy
- ✓ Report the physical montage
- ✓ Preregister a single external test

Limitations: small samples (power 0.84 only if true accuracy were 70%); identical criteria for EEG_2 could not be verified; we tested the paper's best channel-selection configuration, not all.

Aljalal M. et al. Sci Rep 14:12483 (2024) · Kashefpoor M. et al. J Med Signals Sens 6:25 (2016); Biomed Signal Process Control 53:101559 (2019) · Varma & Simon, BMC Bioinformatics 7:91 (2006) · Saeb et al., GigaScience 6:gix019 (2017) · Deb et al., IEEE TEC 6:182 (2002)